MCS-1: Probability and Statistics (MA6.302) Monsoon 2026 Lecture 5

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Machine Learning Lab, IIIT-H

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- Annie is trying to catch a taxi. How many occupied taxis will drive past before she finds one that is taking passengers?
- The number of trials required to get a single success is a **Geometric Random Variable**.

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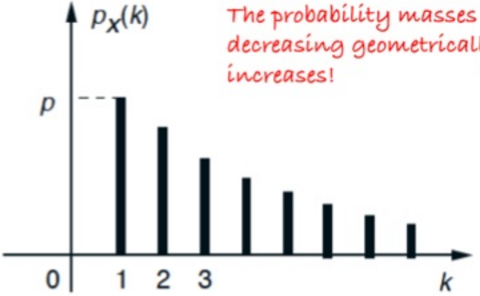

The probability masses are decreasing geometrically as k increases!

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- $X > k$ is the event that $X \geq k+1$, and so $P(X > k) = (1-p)^k$.

Memoryless Property of Geometric Random Variable

What is $P(X > a + b | X > a)$**?**

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$$P(X > a+b|X > a) = \frac{P(X > a+b \text{ and } X > a)}{P(X > a)} = \frac{P(X > a+b)}{P(X > a)}$$
$$= \frac{(1-p)^{a+b)}}{(1-p)^a}$$
$$= (1-p)^b$$
$$= P(X > b)$$

- You forgot about $X > a$ and started the clock afresh!

Poisson Random Variable

I have a book with 10000 words. Probability that a word has a typo is 1/1000. I am interested in how many misprints can be there on average? So a Poisson often shows up when you have a Binomial random variable with very large n and very small p but $n \times p$ is moderate. Here $np = 10$.

Other examples:

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- The number of buses arriving within a given time period.
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We can describe such situations using a **Poisson random variable**.

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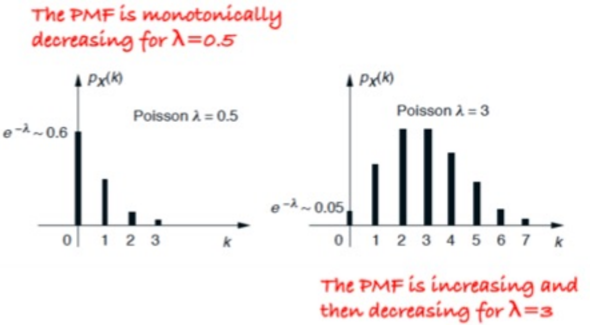

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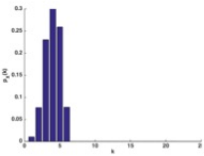
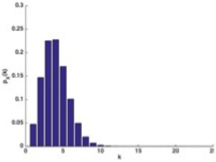


Binomial(5,0.6)



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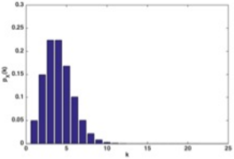
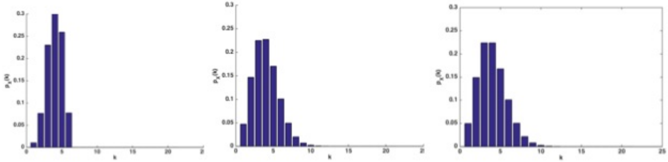


Poisson(3)

Poisson Random Variable



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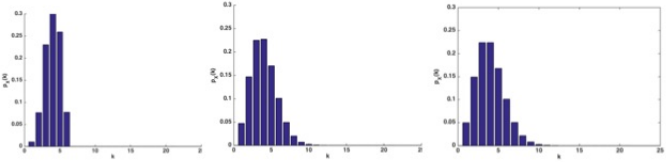

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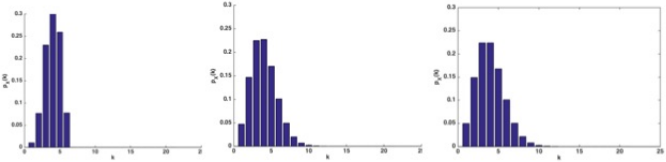

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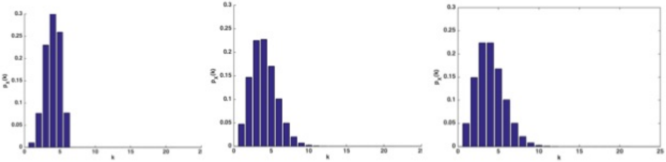
- See how close the PMF's of the $\mathit{Binomial}(100, 0.03)$ and $\mathit{Poisson}(3)$ are!



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- See how close the PMF's of the $Binomial(100, 0.03)$ and $Poisson(3)$ are!
- More formally, we see that ${}^{n}C_k \; p^k(1-p)^{n-k} \approx e^{-\lambda}\frac{\lambda^k}{k!}$ when n is large, k is fixed, and p is small and $\lambda = np$.

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- $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-3} = 0.950.$

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- If you know there is at least one accident, what is the probability that the total number of accidents is at least two?
- $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-3} = 0.950$.
- $P(X \geq 2 | X \geq 1) = P(X \geq 2)/P(X \geq 1) = 0.8/0.950 = 0.84$

Independent Discrete Random Variables

Definition

Two discrete random variables X and Y are said to be independent if

$$P(X = x, Y = y) = P(X = x)P(Y = y), \;\; \forall x \& y$$

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- **Example:** Let X and Y be independent random variables each geometrically distributed with parameter p. Find the distribution of $\min(X, Y)$.

Definition

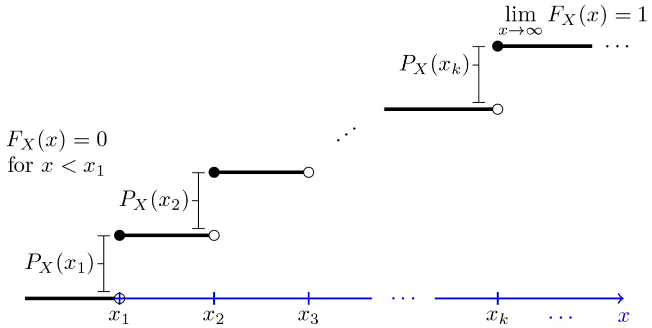
Let X be any random variable, the function

$$F_X(x) = P(X \leq x), \; x \in \mathbb{R}$$

is called CDF function of X.

- **Example:** Toss a coin twice, let X be the number of observed heads. Find the CDF of X.

Let X be a discrete random variable which takes values in the set $\{x_1, x_2, \ldots\}$ such that $x_1 \leq x_2 \leq \ldots$.



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- For $x_i \leq x < x_{i+1}$, we have $F_X(x) = F_X(x_i)$.
- For all $a \leq b$, we have $P(a < x \leq b) = F_X(b) - F_X(a)$.

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  - For example, for $d = 3$ and $n = 5$, if
  - $Y_1 = Y_4 = Y_5 = 2,\ Y_2 = 3$ and $Y_3 = 3$, then $X_1 = 0,\ X_2 = 3$ and $X_3 = 2$

Multinomial Distribution

- We now compute $P(X_1 = x_1, \ldots, X_d = x_d)$ s.t. $x_1 + \ldots + x_d = n$.

$$\begin{aligned} P(X_1 = x_1, \ldots, X_d = x_d) = & P(x_1 \text{ of } (Y_1, \ldots, Y_n) \text{ take value } 1, \\ & x_2 \text{ of } (Y_1, \ldots, Y_n) \text{ take value } 2, \\ & \ldots, \\ & x_d \text{ of } (Y_1, \ldots, Y_n) \text{ take value } d) \end{aligned}$$

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- $Y_1, \ldots, Y_n$ are independent with common density. Thus, every possible outcome of $Y_1, \ldots, Y_n$ with x_1 of them having value 1, x_2 of them having value 2,..., x_d of them having value d, has the same probability, which is $p_1^{x_1} p_2^{x_2} \cdots p_d^{x_d}$.

Multinomial Distribution

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- Let $C(n; x_1, \ldots, x_d)$ denote the number of possible choices, we see that

$$P(X_1 = x_1, \ldots, X_d = x_d) = C(n; x_1, \ldots, x_d)\, p_1^{x_1} p_2^{x_2} \cdots p_d^{x_d}$$

Multinomial Distribution

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- Thus,

$$C(n; x_1, \ldots, x_d) = {}^nC_{x_1} C(n - x_1; x_2, \ldots, x_d) = \frac{n!}{(x_1!)(x_2!)\ldots(x_d!)}$$

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- Since the events $\{X = x_i, Y = z - x_i\}$ are disjoint for different values of i, it follows that

$$P(X + Y = z) = \sum_i P(X = x_i, Y = z - x_i)$$
$$= \sum_i P(X = x_i) P(Y = z - x_i)$$